

Homework 1

- Write a Python program that prints the type of four different literals.
 - Expected output:
The type of this literal is: <class 'bool'>
The type of this literal is: <class 'int'>
The type of this literal is: <class 'str'>
The type of this literal is: <class 'float'>

Homework 2

- Create four different variables using the the valid naming convention and assign to each one a different data type.
 - Expected output:
 - The type of this variable is: <class 'bool'>
 - The type of this variable is: <class 'int'>
 - The type of this variable is: <class 'str'>
 - The type of this variable is: <class 'float'>
- Add a comment to each of the print statements.

Homework 3

- Create one variable for each of the following statements:
 1. Positive integer
 2. Negative integer
 3. Positive float
 4. Negative float
 5. Negative float using scientific notation
- Create a print statements for each of these values and add a comment for each.

Challenge 1

- Write a Python program that allows users to shop in a food items in a market using gold, silver and bronze as currency.
- Objective:
 - Take user input for money in different denominations
 - Convert the total budget to a single unit
 - Let the user select and purchase food items
 - Check affordability, update balances, and repeat until the user exits
- The currency works as follows:
 - 1 Gold Coin = 10 Silver Coins
 - 1 Silver Coin = 100 Bronze Coins
 - 1 Gold Coin = 1,000 Bronze Coins

Challenge 1

- Your program must:
 1. Prompt the user for how many Bronze, Silver, and Gold coins they have (all integers).
 2. Convert their total wealth into a float number of Gold Coins.
 3. Display a menu of food items with float prices in Gold Coins.
 4. Loop:
 1. Ask the user to choose an item by name (string).
 2. Check if the user can afford it.
 - If yes, deduct the price and show the new balance.
 - If no, notify them and let them try again.
 5. Allow the user to type "exit" to end shopping.

Challenge 1

- After exiting, print:
 - A summary of items bought.
 - Final remaining balance in Gold, Silver, Bronze breakdown.
- Each food item is priced in Gold Coins.

- Following is a sample menu

Item	Price (Gold Coins)
-----	-----
Bread	0.75
Cheese	1.50
Apple	0.25
Roast Chicken	3.20
Pie	2.40

Challenge 1 – Sample use

Welcome to the Fantasy Food Market!

Enter Bronze coins: **2**

Enter Silver coins: **2**

Enter Gold coins: **1**

Your total: 1.20 Gold Coins

Menu:

- Bread (0.75 gold)
- Cheese (1.5 gold)
- Apple (0.25 gold)
- Roast Chicken (3.2 gold)
- Pie (2.4 gold)

What would you like to buy? (type 'exit' to finish): **bread**

You bought Bread for 0.75 gold.

Remaining balance: 0.45 gold.

Menu:

- Bread (0.75 gold)
- Cheese (1.5 gold)
- Apple (0.25 gold)
- Roast Chicken (3.2 gold)
- Pie (2.4 gold)

What would you like to buy? (type 'exit' to finish): **pie**

You cannot afford Pie. You have 0.45 gold.

Menu:

- Bread (0.75 gold)
- Cheese (1.5 gold)
- Apple (0.25 gold)
- Roast Chicken (3.2 gold)
- Pie (2.4 gold)

What would you like to buy? (type 'exit' to finish): **exit**

--- Shopping Summary ---

Items purchased: Bread

Final balance: 0.45 gold → 0 gold, 4 silver, 50 bronze

Thank you for visiting!